

PHYSICS**Paper 1 (Theory)****Maximum Marks: 70****Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional fifteen minutes for only reading the question paper.
2. You must NOT start writing during reading time.
3. This question paper has 9 printed pages.
4. It has twenty questions in this paper. Answer all questions.
5. There are four sections in the paper: A, B, C and D. Internal choices have been provided in two questions each in Sections B, C and D.
6. Section A consists of one question having fourteen subparts of one mark each.
7. While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
8. Section B consists of seven questions of two marks each.
9. Section C consists of nine questions of three marks each.
10. Section D consists of three questions of five marks each.
11. The intended marks for questions are given in brackets [].
12. A list of useful constants and relations is given below these instructions.
13. A simple scientific calculator without a programmable memory may be used for calculations.

USEFUL CONSTANTS AND RELATIONS

1.	Planck's constant	h	=	$6.6 \times 10^{-34} \text{ J s}$
2.	Speed of light in vacuum	c	=	$3 \times 10^8 \text{ m s}^{-1}$
3.	Charge of a proton	e	=	$1.6 \times 10^{-19} \text{ C}$
4.	Mass of an electron	m_e	=	$9.1 \times 10^{-31} \text{ kg}$
5.	Permittivity of vacuum	ϵ_0	=	$8.85 \times 10^{-12} \text{ F m}^{-1}$
6.		$1/4\pi\epsilon_0$	=	$9 \times 10^9 \text{ N m}^2 \text{ C}^{-2}$
7.	Permeability of vacuum	μ_0	=	$4\pi \times 10^{-7} \text{ H m}^{-1}$
8.	One atomic mass unit	1 u	=	$1.66 \times 10^{-27} \text{ kg} = 931.5 \text{ MeV}/c^2$
9.	One electron volt	1 eV	=	$1.6 \times 10^{-19} \text{ J}$
10.		1 MeV	=	$1.6 \times 10^{-13} \text{ J}$
11.	Rydberg constant	R	=	$1.097 \times 10^7 \text{ m}^{-1}$

Note: The questions in this sample paper follow the format of the ISC Specimen Question Paper 2027. The weightage allocated to each unit in the syllabus has been strictly adhered to.

SECTION A – 14 MARKS

Question 1

(A) In questions (i) to (vii), choose the correct alternative (a), (b), (c) or (d) for each of the questions given below.

(i) Identify the correct statements about equipotential surfaces. (*Understand*) [1]

(P) The electric field at every point is perpendicular to the equipotential surface through it.

(Q) No work is done in moving a charge along an equipotential surface.

(R) Two equipotential surfaces can intersect each other.

(S) Equipotential surfaces are closer together where the field is weaker.

(a) Both (P) and (Q)

(b) Both (Q) and (R)

(c) Both (R) and (S)

(d) Both (P) and (S)

(ii) A material has a relative permeability of 0.99995. The material is: (*Understand*) [1]

(a) diamagnetic

(b) paramagnetic

(c) ferromagnetic

(d) a superconductor below its critical temperature

(iii) An electron beam passes undeflected through mutually perpendicular electric and magnetic fields of $3 \times 10^4 \text{ V m}^{-1}$ and $2 \times 10^{-3} \text{ T}$. The speed of the electrons is: (*Apply*) [1]

(a) $6 \times 10^1 \text{ m s}^{-1}$

(b) $1.5 \times 10^7 \text{ m s}^{-1}$

(c) $6.7 \times 10^{-8} \text{ m s}^{-1}$

(d) $1.5 \times 10^4 \text{ m s}^{-1}$

(iv) Match the entries in Column I with those in Column II. (*Recall*) [1]

Column I	Column II
1. Work function	A. h/p
2. Stopping potential	B. Minimum energy needed to eject an electron from the metal surface
3. Threshold frequency	C. Negative anode potential at which the photocurrent just becomes zero
4. de Broglie wavelength	D. Minimum frequency of light that can cause photoemission

(a) 1-B, 2-C, 3-D, 4-A

(b) 1-C, 2-B, 3-D, 4-A

(c) 1-B, 2-D, 3-C, 4-A

(d) 1-A, 2-C, 3-D, 4-B

- (v) Select the odd one out from the following dopants. (*Analyse*) [1]
- (a) Boron
 - (b) Indium
 - (c) Gallium
 - (d) Arsenic
- (vi) Given below are two statements marked, Assertion and Reason. Read the two statements and choose the correct option. [1]
- Assertion:** A biconvex glass lens of refractive index 1.5 becomes practically invisible and has no focusing effect when immersed in a liquid of refractive index 1.5.
- Reason:** The power of a lens is proportional to $(n_{\text{lens}}/n_{\text{medium}} - 1)$. (*Analyse*)
- (a) Both Assertion and Reason are true, and Reason is the correct explanation for Assertion.
 - (b) Both Assertion and Reason are true, but Reason is not the correct explanation for Assertion.
 - (c) Assertion is true and Reason is false.
 - (d) Assertion is false and Reason is true.
- (vii) Two coherent sources whose intensities are in the ratio 9 : 1 produce an interference pattern. The ratio of the maximum to the minimum intensity is: (*Apply*) [1]
- (a) 9 : 1
 - (b) 3 : 1
 - (c) 4 : 1
 - (d) 16 : 1
- (B) Answer the following questions briefly:
- (i) A uniform wire of resistance $16\ \Omega$ is bent into a circle. What is the resistance between two diametrically opposite points? (*Apply*) [1]
 - (ii) What is the average power consumed over a complete cycle by a pure inductor connected to an a.c. source? (*Recall*) [1]
 - (iii) Fill in the blank with an appropriate term from the bracket.
In single-slit diffraction, the central maximum is _____ each secondary maximum. (half as wide as, as wide as, twice as wide as) (*Recall*) [1]
 - (iv) State two conditions necessary for total internal reflection. (*Recall*) [1]
 - (v) Calculate the de Broglie wavelength of an electron moving with a speed of $1.1 \times 10^6\ \text{m s}^{-1}$. (*Apply*) [1]
 - (vi) An electron and a positron at rest annihilate each other. Why must at least two photons be produced? (*Understand*) [1]
 - (vii) What is meant by the forbidden energy gap of a semiconductor? (*Recall*) [1]

SECTION B – 14 MARKS

Question 2

- (i) A point charge of $8.85 \mu\text{C}$ is placed at the centre of a cube of side 10 cm. Calculate the electric flux through (a) the whole cube and (b) one face of the cube. (*Apply*) [2]

OR

- (ii) Two point charges $+3 \text{ nC}$ and -3 nC are 2 mm apart. Find the dipole moment. Calculate the torque on this dipole in a uniform field of $5 \times 10^4 \text{ N C}^{-1}$ when its axis makes 30° with the field. (*Apply*) [2]

Question 3

- Each key of a computer keyboard is a small parallel plate capacitor of plate area 1.0 cm^2 with an air gap of 0.5 mm. Pressing the key reduces the gap to 0.15 mm. Calculate the change in capacitance that the circuit detects. (*Apply*) [2]

Question 4

- (i) Figure 1 shows the I–V graphs of the same metallic wire at two temperatures T_1 and T_2 . Which temperature is higher? Give a reason. (*Analyse*) [2]

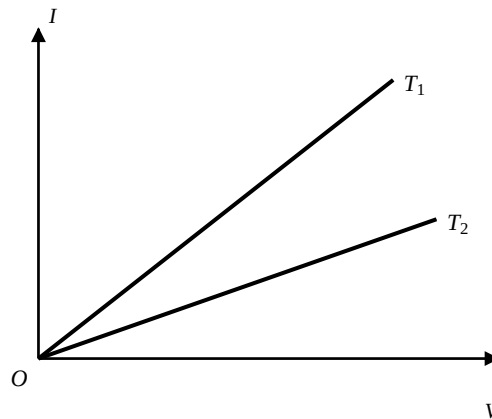


Figure 1

OR

- (ii) A copper coil has a resistance of 20Ω at 20°C . The temperature coefficient of resistance of copper is $4.0 \times 10^{-3} ^\circ\text{C}^{-1}$. Find its resistance at 70°C . (*Apply*) [2]

Question 5

- Two moving-coil galvanometers G_1 and G_2 have the same coil area, magnetic field and suspension. G_1 has 60 turns and resistance 12Ω ; G_2 has 30 turns and resistance 4Ω . Which has the greater (a) current sensitivity and (b) voltage sensitivity? Justify. (*Analyse*) [2]

Question 6

- A thin equiconvex lens of power $+5 \text{ D}$ is cut into two identical plano-convex halves by a plane perpendicular to its principal axis. (a) What is the power of each half? (b) If the two halves are put back together with their plane faces in contact, what is the power of the combination? (*Analyse*) [2]

Question 7

- (i) Identify the electromagnetic radiation (a) used in night-vision cameras, and (b) of wavelength about 10^{-12} m emitted by nuclei. (*Recall*) [1]
- (ii) State two properties common to all electromagnetic waves. (*Recall*) [1]

Question 8

- Light of frequency ν ejects photoelectrons of maximum kinetic energy 1.0 eV from a metal of work function 2.0 eV. (a) What happens to the maximum kinetic energy if the intensity is doubled? (b) Find the maximum kinetic energy if the frequency is doubled. (*Analyse*) [2]

SECTION C – 27 MARKS**Question 9**

- (i) Define electric potential at a point. Derive an expression for the electric potential at a distance r from a point charge $+q$ in vacuum. (*Understand*) [3]

OR

- (ii) Obtain an expression for the electric field at a point on the axial line of a short electric dipole of moment p , at a distance r from its centre. Hence show that at the same distance the axial field is twice the equatorial field ($E_{eq} = p/4\pi\epsilon_0 r^3$). (*Understand*) [3]

Question 10

- (i) In a potentiometer experiment to find the internal resistance of a cell, the balance length is 150 cm with the cell on open circuit. When a resistance of $5\ \Omega$ is connected across the cell, the balance length becomes 125 cm. (a) Calculate the internal resistance of the cell. (b) How would the balance lengths change if the current in the potentiometer wire were increased? (*Analyse*) [3]

OR

- (ii) In the network of Figure 2 the cells have negligible internal resistance. Using Kirchhoff's laws, find the currents I_1 , I_2 and I_3 . (*Apply*) [3]

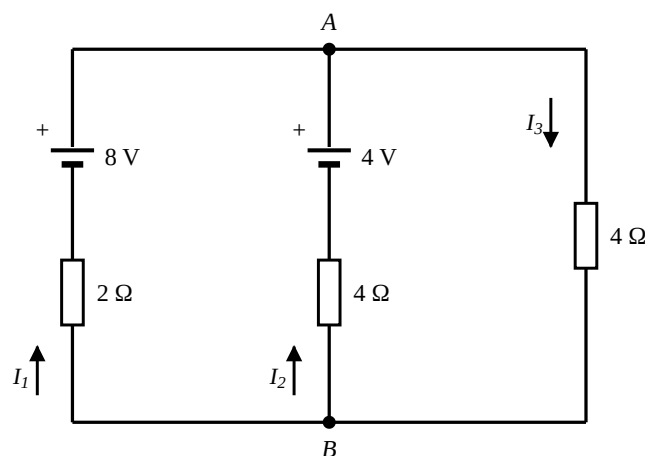


Figure 2

Question 11

State Ampère's circuital law. Using it, obtain an expression for the magnetic field at a distance r from a long straight wire carrying a current I . (*Understand*) [3]

Question 12

- (i) A galvanometer of resistance $50\ \Omega$ gives full-scale deflection at 4 mA . Find the shunt needed to convert it into an ammeter of range $0\text{--}2\text{ A}$, and the resistance of the ammeter. (*Apply*) [2]
- (ii) Why is the core of a transformer laminated? (*Understand*) [1]

Question 13

Using the arrangement of Young's double slit experiment shown in Figure 3 (monochromatic light of wavelength 600 nm), a teacher asked the following questions. State whether the answer of the student is correct or incorrect. Correct the incorrect response with a brief explanation. (*Evaluate*)

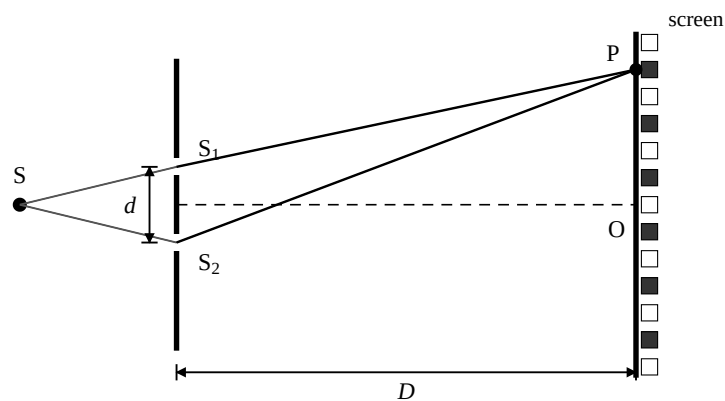


Figure 3

- (i) **Teacher:** At P the path difference is $1.5 \times 10^{-6}\text{ m}$. Is P bright or dark? [1]
Student: Dark, because the path difference is an odd multiple of $\lambda/2$.
- (ii) **Teacher:** What happens to the fringe width if the screen is moved to twice its distance from the slits? [1]
Student: The fringe width is halved.
- (iii) **Teacher:** What will be seen if white light is used? [1]
Student: All the fringes, including the central one, will be coloured.

Question 14

A plane wavefront travelling in glass is incident obliquely on a plane glass–air boundary (the angle of incidence is less than the critical angle). Using Huygens' principle, draw the refracted wavefront and hence prove Snell's law. (*Analyse*) [3]

Question 15

- (i) Draw a graph showing how the angle of deviation δ produced by a prism varies with the angle of incidence i . Mark the angle of minimum deviation. (*Understand*) [1]
- (ii) A thin prism of angle 5° has refractive indices 1.66 for violet and 1.62 for red light. Calculate the angular dispersion and the dispersive power (take the mean refractive index as 1.64). (*Apply*) [2]

Question 16

- (i) Draw a labelled ray diagram of a compound microscope forming its final image at the least distance of distinct vision. (*Understand*) [1]
- (ii) A compound microscope has an objective of focal length 1 cm and an eyepiece of focal length 5 cm. An object is placed 1.1 cm from the objective and the final image is formed at 25 cm. Calculate the magnifying power and the separation of the lenses. (*Apply*) [2]

Question 17

Figure 4 shows the variation of stopping potential V_s with the frequency ν of the light incident on a metal surface.

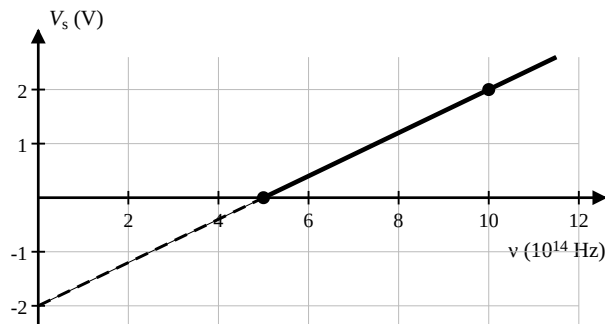


Figure 4

- (i) What is the threshold frequency of the metal? (*Analyse*) [1]
- (ii) Use the graph to find Planck's constant. (*Apply*) [1]
- (iii) What is the work function of the metal in eV? (*Apply*) [1]

SECTION D – 15 MARKS

Question 18

- (i)(a) An alternating current $i = 5\sqrt{2} \sin(120\pi t)$ A flows through a heater of resistance 40Ω . Find (1) the rms current, (2) the frequency, and (3) the average power consumed and the rms voltage across the heater. (*Apply*) [3]

- (b) A square coil of side 10 cm, 20 turns and resistance $2\ \Omega$ is held with its plane perpendicular to a magnetic field which varies as shown in Figure 5. Find the induced current during 0–0.1 s, 0.1–0.3 s and 0.3–0.5 s. (*Apply*) [2]

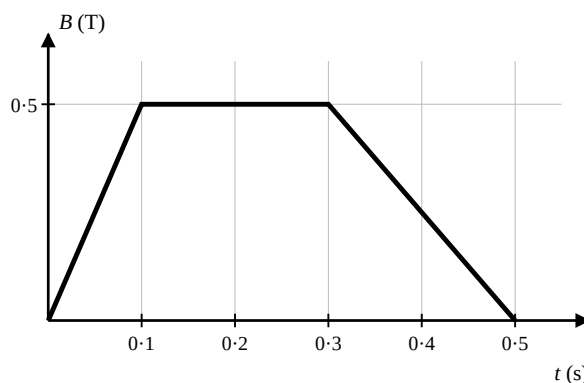


Figure 5

OR

- (ii) A 220 V, 50 Hz source is connected to a resistor of $30\ \Omega$ in series with an unknown element X. The rms current is 4.4 A and it **leads** the source voltage.
- Identify X. Give a reason. (*Analyse*) [1]
 - Find the impedance and the reactance of X. (*Apply*) [1]
 - Find the capacitance / inductance of X. (*Apply*) [1]
 - Find the power factor and the average power. (*Apply*) [1]
 - Draw the phasor diagram for the circuit. (*Understand*) [1]

Question 19

- An α -particle of kinetic energy K is projected head-on towards a nucleus of atomic number Z . Obtain an expression for its distance of closest approach. Calculate it for a 5 MeV α -particle and a gold nucleus ($Z = 79$). (*Apply*) [3]
- Calculate the shortest wavelength of the Balmer series of hydrogen. (*Apply*) [1]
- In which region of the electromagnetic spectrum does the Lyman series lie? (*Recall*) [1]

OR

- Using de Broglie's hypothesis, show that the angular momentum of an electron in a stationary orbit is an integral multiple of $h/2\pi$. (*Apply*) [2]
- Calculate the mass defect, the binding energy and the binding energy per nucleon of ${}^{40}_{20}\text{Ca}$. (Mass of ${}^{40}\text{Ca}$ atom = 39.962591 u, mass of ${}^1\text{H}$ atom = 1.007825 u, mass of neutron = 1.008665 u) (*Apply*) [3]

Question 20

Riya studies a silicon junction diode in the laboratory. She first plots its forward characteristic (Figure 6) using a milliammeter, then measures the reverse current with a microammeter. Next she uses the diode to rectify the 50 Hz mains stepped down by a transformer. Finally she builds the circuit of Figure 7, in which each diode has a forward resistance of $30\ \Omega$ and an infinite reverse resistance.

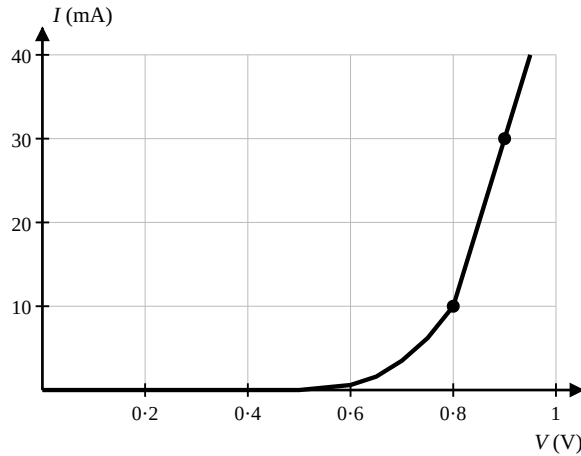


Figure 6

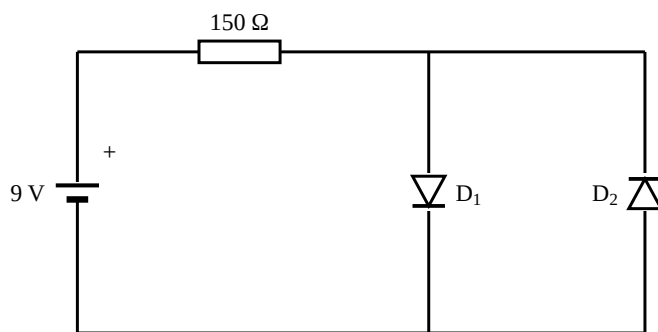


Figure 7

- (i) Find the dynamic resistance of the diode between 0.8 V and 0.9 V. (*Apply*) [1]
- (ii) Why does she need a microammeter for the reverse characteristic? (*Understand*) [1]
- (iii) Draw the circuit of a half-wave rectifier and show its input and output waveforms. (*Understand*) [1]
- (iv) Name the diode that is designed to operate in the reverse breakdown region, and state its use. (*Recall*) [1]
- (v) Find the current drawn from the 9 V battery in Figure 7. (*Apply*) [1]

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